

## Environment Description:

My environment used two virtual machines, each VM was running the current version of Windows 11 pro. (StudentVM) configured with firewall and all options enabled in windows defender anti-virus. (StudentNoAV) configured with firewall and all windows defender anti-virus options disabled. I added both endpoints to LimaCharlie EDR to monitor alerts and responses.

## Methodology:

### Scenario A – EICAR Detection Test

- Each machine was tested with the EICAR test file, I pasted the EICAR string into notepad saved it as eicar.com and selected (All file types). The file was double clicked to attempt execution, and behavior was monitored on both machines and LimaCharlie Management Console.

### Scenario B - Scripted Activity (PowerShell / LOLBins)

- The following commands were ran from an administrator powershell on each device and results were monitored on both machines and in the LimaCharlie Management Console.
  - Get-Process | Select-Object -First 20
  - Invoke-WebRequest -Uri https://example.com
  - powershell.exe -NoP -Exec Bypass -C "whoami; Get-Process | Select-Object -First 10"
  - certutil -urlcache -split -f https://example.com C:\Users\Public\test.html

After each scenario I monitored the following sections in the LimaCharlie GUI.

- Detections page (filtered by endpoint)
- Timeline view (for process execution chains, network connections, and file writing)
- Parent/child process tree

## Findings:

### Scenario A:

I am giving you a side by side table comparison of what I observed on both machines during the EICAR test.

| Aspect                             | StudentVM (Defender ON)                   | StudentNoAV (Defender OFF)                           | Key Observation                                 |
|------------------------------------|-------------------------------------------|------------------------------------------------------|-------------------------------------------------|
| EICAR File Creation                | Blocked / Quarantined by Windows Defender | File successfully created (eicar.com)                | Defender provided strong prevention             |
| File Execution Attempt             | Prevented by Defender                     | File existed but did not execute (16-bit limitation) | Defender stopped at file level                  |
| Windows Defender Reaction          | Clear quarantine notification + action    | No action / no notification                          | Defender Blocked the creation and quarantined.  |
| LimaCharlie Detections             | No detections triggered                   | No detections triggered                              | EDR did not flag EICAR                          |
| LimaCharlie Telemetry / Events     | Limited (file blocked early)              | File write event likely logged in Timeline           | Minimal behavioral telemetry due to early block |
| Overall Protection                 | Strong (prevented threat)                 | None (file persisted)                                | Clear advantage with Defender enabled           |
| Forensic Visibility in LimaCharlie | Low                                       | Moderate                                             | EDR visibility depends on prevention layer      |

Protection history

View the latest protection actions and recommendations from Windows Security.

All recent items Filters

Sign in with Microsoft for enhanced security and other...  
5/10/2026 2:12 PM

Set up OneDrive for file recovery options in case of a ra...  
5/10/2026 2:12 PM

The settings to block potentially unwanted apps are tur...  
5/10/2026 2:12 PM

**Threat quarantined**  
5/10/2026 2:11 PM Severe

Detected: Virus:DOS/EICAR\_Test\_File  
Status: Quarantined  
Quarantined files are in a restricted area where they can't harm your device. They will be removed automatically.

Date: 5/10/2026 2:11 PM  
Details: This program is dangerous and replicates by infecting other files.

Affected items:

file: C:\Users\Student\Documents\eicar.com

[Learn more](#)

Actions

EICAR quarantined by Windows AV

EICAR detected upon saving

EICAR Existing after saving with no AV

## Scenario B:

This is my side by side comparison during the scripted activity test

| Test Activity                                                | StudentVM (Defender ON)                                                           | StudentNoAV (Defender OFF)                                                        | Key Observation                                                            |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>PowerShell Reconnaissance (whoami, Get-Process)</b>       | LimaCharlie detections: Local Accounts Discovery, Non-Interactive PowerShell      | Same detections: Local Accounts Discovery, Non-Interactive PowerShell             | Strong behavioral detection on <b>both</b> endpoints                       |
| <b>Suspicious PowerShell Parameters (-NoP - Exec Bypass)</b> | Detected ("Suspicious PowerShell Parameter Substring") Change PowerShell Policies | Detected ("Suspicious PowerShell Parameter Substring") Change PowerShell Policies | EDR consistently flags dangerous flags                                     |
| <b>Invoke-WebRequest (Network Connection)</b>                | Logged in Timeline (PowerShell network connection)                                | Logged in Timeline (PowerShell network connection)                                | Not blocked by Defender; fully visible on both                             |
| <b>Certutil Download (LOLBin)</b>                            | Blocked by Windows Defender                                                       | File downloaded + "Suspicious Download Via Certutil.EXE" detection                | Defender provided prevention, EDR didn't prevent.                          |
| <b>Process Execution Chain</b>                               | Visible in Timeline                                                               | Clear visible chain (cmd.exe → certutil.exe, PowerShell)                          | Excellent visibility                                                       |
| <b>File Write Events</b>                                     | Blocked early by Defender                                                         | test.html successfully created and logged                                         | Clear prevention difference                                                |
| <b>Automated Response Actions</b>                            | Didn't let command execute                                                        | None (Detection / Alert only)                                                     | Blocked by Windows AV, Not by EDR                                          |
| <b>Overall LimaCharlie Visibility</b>                        | Good behavioral visibility                                                        | High behavioral visibility                                                        | EDR Shows more Logs with NoAV. Events blocked by AV aren't visible in EDR. |

Ran the same LOLBin command on each machine, one was blocked by AV one wasn't

```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>certutil -urlcache -split -f https://example.com C:\Users\Public\test.html
Access is denied.

C:\Windows\System32>

Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>certutil -urlcache -split -f https://example.com C:\Users\Public\test.html
**** Online ****
0000 ...
0210
CertUtil: -URLCache command completed successfully.
  
```

## Detections

[VIEW DOCS →](#)

| SOURCE                | DATE                | RANGE                |
|-----------------------|---------------------|----------------------|
| studentvm.localdomain | 2026-05-11 00:48:18 | <input type="text"/> |

Quick Search

Add

You're up-

2026-05-10 21:55:41 Local Accounts Discovery → studentvm.localdomain {"event":{"COMMAND\_LINE":"\\C:\\WINDOWS'  
2026-05-10 21:55:40 Non Interactive PowerShell Process Spawned → studentvm.localdomain {"event":{"COMMAND\_LI  
2026-05-10 21:55:40 Change PowerShell Policies to an Insecure Level → studentvm.localdomain {"event":{"COMMA  
2026-05-10 21:55:40 Suspicious PowerShell Parameter Substring → studentvm.localdomain {"event":{"COMMAND\_LIN  
2026-05-10 21:30:58 Non Interactive PowerShell Process Spawned → studentvm.localdomain {"event":{"BASE\_ADDRE  
2026-05-10 21:01:00 Suspicious Process Masquerading As SvcHost.EXE → studentvm.localdomain {"event":{"FILE\_I  
2026-05-10 21:01:00 Suspicious Process Masquerading As SvcHost.EXE → studentvm.localdomain {"event":{"FILE\_I  
2026-05-10 21:00:59 Suspicious Process Masquerading As SvcHost.EXE → studentvm.localdomain {"event":{"FILE\_I  
2026-05-10 21:00:56 Suspicious Process Masquerading As SvcHost.EXE → studentvm.localdomain {"event":{"FILE\_I

All Detection  
events by  
LimaCharlie on  
StudentVM

## Detections

[VIEW DOCS →](#)

| SOURCE                  | DATE                | RANGE                |
|-------------------------|---------------------|----------------------|
| studentnoav.localdomain | 2026-05-11 00:49:13 | <input type="text"/> |

Quick Search

You'r

2026-05-10 22:12:43 Local Accounts Discovery → studentnoav.localdomain {"event":{"COMMAND\_LINE":"\\C:\\W  
2026-05-10 22:12:43 Change PowerShell Policies to an Insecure Level → studentnoav.localdomain {"event":{"  
2026-05-10 22:12:43 Suspicious PowerShell Parameter Substring → studentnoav.localdomain {"event":{"COMMA  
2026-05-10 22:12:43 Non Interactive PowerShell Process Spawned → studentnoav.localdomain {"event":{"COMM  
2026-05-10 22:02:24 00403-WIN-Certutil\_Network\_Connection\_to\_Public\_Address → studentnoav.localdomain {"  
2026-05-10 22:02:18 00040-WIN-CertUtil\_File\_Download\_urlcache → studentnoav.localdomain {"event":{"BASE\_  
2026-05-10 22:02:18 Suspicious Download Via Certutil.EXE → studentnoav.localdomain {"event":{"BASE\_ADDRE  
2026-05-10 21:27:11 Non Interactive PowerShell Process Spawned → studentnoav.localdomain {"event":{"BASE  
2026-05-10 20:40:50 Non Interactive PowerShell Process Spawned → studentnoav.localdomain {"event":{"BASE  
2026-05-10 20:40:48 Suspicious Process Masquerading As SvcHost.EXE → studentnoav.localdomain {"event":{"  
2026-05-10 20:40:47 Suspicious Process Masquerading As SvcHost.EXE → studentnoav.localdomain {"event":{"  
2026-05-10 20:40:47 Suspicious Process Masquerading As SvcHost.EXE → studentnoav.localdomain {"event":{"  
2026-05-10 20:40:47 Suspicious Process Masquerading As SvcHost.EXE → studentnoav.localdomain {"event":{"

All Detection  
events by  
LimaCharlie on  
StudentNoAV



Threat blocked  
5/10/2026 3:00 PM

Severe ^

Detected: Trojan:Win32/Ceprolad.A  
Status: Removed  
A threat or app was removed from this device.

Date: 5/10/2026 3:00 PM  
Details: This program is dangerous and executes commands from an attacker.

**Affected items:**

CmdLine: C:\Windows\System32\certutil.exe -urlcache -split -f https://  
example.com C:\Users\Public\test.html

[Learn more](#)

Actions ▾

Threat was blocked by  
Windows AV from the  
LOLBin command.

## **1. How did Windows Defender's presence change the outcome and logs?**

Windows Defender's presence seemed to change the logs by not logging the events that windows defender had already taken care of. When Windows Defender wasn't present, it seemed that there were more logs because less things were being blocked at the endpoint, therefore all the telemetry would get sent to the EDR alerting you based on your policy.

## **2. What did OpenEDR add beyond native AV?**

The original OpenEDR installation produced very limited alerts and telemetry — it mostly just logged file-write events. Because of that, I switched to LimaCharlie EDR for the lab. LimaCharlie provided an abundance of telemetry including process creation, network connections, PowerShell commands etc. It was a detailed log of all events happening on your endpoint, providing you with process tree execution chains and alerts to events they deemed to be suspicious. You can also send direct commands to the endpoint from the management console.

### **Recommendations:**

Some gaps that I observed when it comes to EDR with AV and without AV were that the EDR didn't log the events already blocked by the AV. Although, without the AV the EICAR file was able to remain on the computer without the EDR quarantining it. If we used a more advanced EDR with a paid for license, I'm sure all these events would be blocked and taken care of depending on your policies. As far as system hardening goes, I would suggest keeping windows AV active because it acts as a line of defense before telemetry is sent to the EDR. Having the AV disabled to generate extra telemetry wouldn't be worth leaving your endpoints vulnerable to attack.